

顏子傑 (YAN, ZIH-JIE)

Updated October 3, 2026

Research interests

Convectively Coupled Kelvin Waves, Numerical Modeling,
Moist Convection, Machine Learning.

Education

National Taiwan University (NTU) Taipei, Taiwan
Undergraduate student in Atmospheric Science 2023/09 – Present

Research experience

Undergraduate Summer Research Program

Longwave Moisture-Radiation Feedback and Kelvin Wave Modulation.

Mentor: Professor Kai-Chih Tseng Summer 2025

9th CHiA Forum Poster Presentation

Sensitivity of GMS to Convective Entrainment: Dynamics or Thermodynamics?

Mentor: Professor Jenny Summer 2026

Research Assistant

Mentor: Professor Jenny 2025/09 – Present

Others

Contributed to model development. Source code: [idealized-spectral-gem](#).

Selected Projects

Web and Collaborative Software Development

Participated in a software engineering project for the Data Structures and Algorithms course in CSIE, Built an interactive algorithm visualization website.

Source code: [JackQuark/SoftwareEngineering](#)

3DVar Implementation based on Lorenz-96 Model

Implemented a 3D-Variational (3DVar) data assimilation system based on the Lorenz-96 mathematical model.

Source code: [JackQuark/DA2026_FinalProject](#)

Skills

Programming & Tools

Languages: Python, C, C++, C#, Fortran, Julia, GNU Bash.

Tools & Environments: Linux, Git (Version Control), Unity.

Modeling & Software Engineering

Extensive experience in source code optimization and modification for numerical models (Julia-based Dycore, GFDL Isca).

Familiar with machine learning, software development collaboration, and agent development workflow.